2044 Machine learning prediction of perineural invasion in hepatobiliary cancers

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Introduction

Hepatobiliary cancers are projected to double in incidence by 2029, ranking 6th globally in cancer burden.

Perineural invasion (PNI) is a key pathological feature where tumor cells invade in, around, or through nerves and their sheaths.

PNI is strongly associated with tumor aggressiveness, metastatic potential, and poor prognosis.

Previous research has primarily: Focused on PNI in non-hepatobiliary cancers, like gastric and pancreatic cancers.

Machine Learning (ML) has shown promise in predicting invasive features using clinical and imaging data.However, no comprehensive ML models currently exist for PNI prediction in hepatobiliary cancers.

The study aims to: (1) identify clinicopathologic, genomic, and demographic predictors of PNI in hepatobiliary cancers. (2) develop machine learning classification models that integrate multi-modal data for preoperative PNI prediction.

Methodology

Data Source: MSK HCC (2024) via cBioPortal; includes clinical, pathological, and genomic profiles (N = 434).

Data Processing: Cleaned and encoded variables (binary, label, and one-hot encoding).

Variable Types:

- **Pre-operative**: Age, BMI, TMB, MSI, Tumor Stage, Purity, Comorbidities, Imaging, Mutations, Node status, multifocality.
- **Peri-/Intra-operative**: Resection type, sample, margin.Post-op: Systemic therapies

Split into 80% training / 20% testing sets.

Sub-Analyses (N = 253):

- Separate models for IHC and EHC cholangiocarcinomas.
- Included mutation data: BAP1, TP53, IDH1, KRAS, SMAD4.
- Built preoperative-only models for both IHC and EHC.

Statistical Analysis: Descriptive stats, logistic regression, ORs, and p-values (<0.05).

ML Models Used: Logistic Regression (LR), SVM, KNN, Decision Trees, Random Forest (RF), Gradient Boosting (GB), Adaboost, Guassian NB, MLP.

GridSearchCV with nested cross-validation for hyperparameter tuning.

Evaluated with Accuracy, F1, AUROC, etc.

Explainability: Applied SHAP analysis to interpret feature impact on PNI predictions.

Figure 1: Area Under the Curve

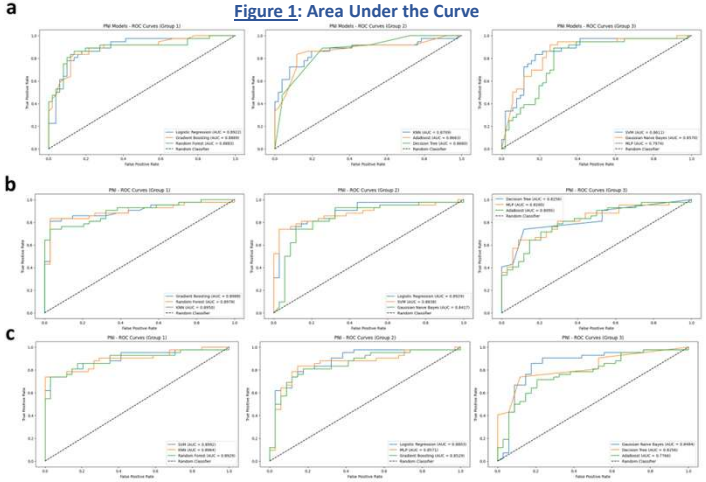


Figure 3: Cholangiocarcinoma Cohort

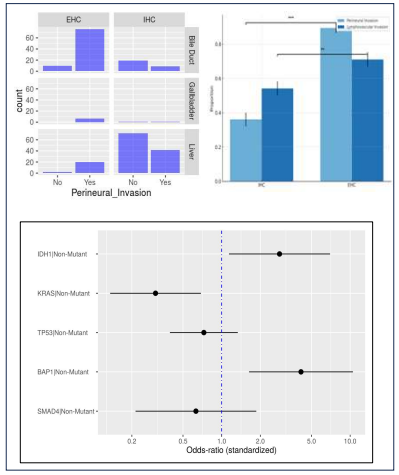


Figure 2: SHAP

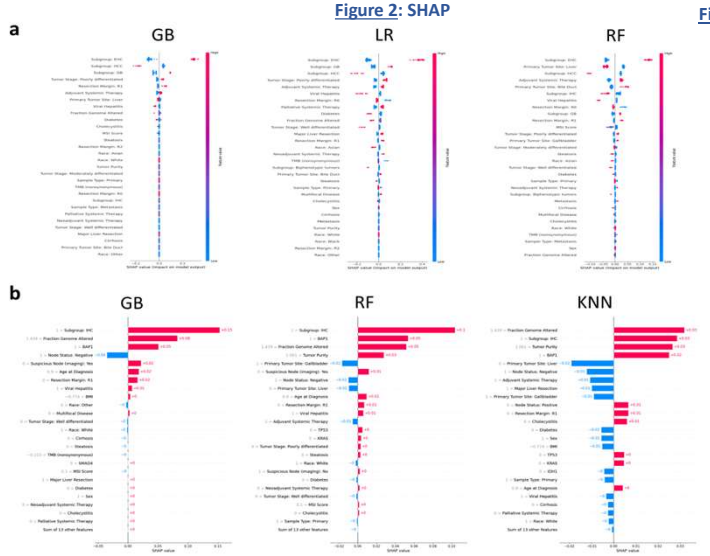
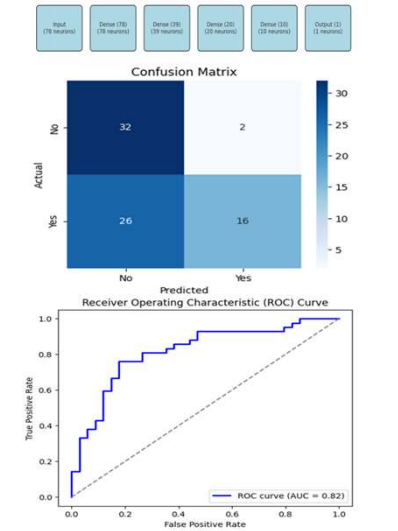


Figure 4: Cholangiocarcinoma – TensorFlow MLP



Results

Figure 1: AUROC values for (a) PNI prediction using all variables and all hepatobiliary cancers, (b) PNI prediction on Cholangiocarcinoma cohort using all variables, and (c) PNI prediction on Cholangiocarcinoma cohort using pre-operative variables.

Best Models: *LR, GB, RF, KNN, SVM*

Figure 2: SHAP Value Analysis of top-performing models for (a) PNI in all hepatobiliary cancers, and (b) PNI in Cholangiocarcinoma.

Most Influential Variables:

All hepatobiliary cancers (All Variables): *Subgroup, Tumor Stage*

Cholangiocarcinoma (All Variables): *Fraction genome altered, BAP1, subggroup*

Figure 3: Distribution of EHC and IHC across PNI status (top left). Difference in PNI proportion across IHC and EHC (top right). Binary logistic regression for PNI status (dependent variable) utilizing mutational status predictors.

PNI in EHC > IHC (significant)

BAP1, IDH1, and KRAS mutations are significant predictors of PNI

Figure 4: Performance of TensorFlow MLP for PNI prediction in Cholangiocarcinoma cohort. MLP architecture (top), confusion matrix (middle), and ROC curve (bottom).

Conclusion

ML models can achieve good performance in predicting PNI in hepatobiliary cancers.

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